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Computational and Statistical Guarantees of the c -Rectified Flow

A Cost-Aware Flow Matching Variant Provably Minimises User-Specified Transport Costs While Preserving Marginals, With Sample Complexity Bounds

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Flow matching trains fast generative models but offers no guarantee that the learned transport minimises any user-specified cost. The *c-Rectified Flow* restricts learned vector fields to gradient-of-convex-conjugate form and proves that iterative application monotonically reduces c -transport cost while automatically preserving marginal distributions—with sample complexity bounds characterising how many data points are needed.

AT A GLANCE

Problem: Standard rectified flow / flow matching learns a valid generative model but provides no guarantee of minimising any user-specified transport cost.

Why it matters: Many scientific applications (molecule generation, image editing, optimal control) have explicit cost functions; a theoretically grounded cost-aware flow model is essential for safe deployment.

Method: Restrict vector fields to $\nabla \bar{c}(\nabla f_t(\cdot))$ form; iterative application provably reduces c -transport cost.

Result: Computational and statistical guarantees for cost-aware flow learning; conditions for recovery of the true optimal transport solution.

Evidence: Formal proofs of convergence and sample complexity; counterexamples delineating when strong assumptions are necessary.

The Big Picture

Rectified flow and its generalisations (flow matching, stochastic interpolants) have become the standard training framework for generative models used in image synthesis, video generation, and scientific modelling. The core idea is simple: train a neural ODE velocity field $v_t(x)$ to push samples from a source distribution π_0 (e.g., Gaussian noise) towards a target distribution π_1 (e.g., data), with training implemented as a simple regression objective.

This works remarkably well in practice. But for applications where the *how* of transport matters—molecule generation (minimise conformational energy), image editing (preserve perceptual structure), optimal control (minimise actuator effort)—standard flow matching provides no guarantee that the learned transport is cost-efficient. It may find a valid coupling $\pi_0 \rightarrow \pi_1$ that is wasteful or even physically unrealisable.

c -Rectified Flow closes this gap.

Why Existing Approaches Fall Short

Standard rectified flow minimises the squared- L_2 distance between a straight-line trajectory and the learned trajectory, which corresponds to the L_2 transport cost. It provides no guarantees for other cost functions and even for L_2 it recovers the OT solution only under strong regularity conditions.

Iterative rectification (reflow) applies rectified flow recursively on coupled samples to straighten trajectories, reducing L_2 cost in practice but providing no monotone convergence guarantee for general costs c .

Minibatch OT pre-computes approximate couplings before training, but the approximation error accumulates and there is no learning-theoretic analysis of the resulting flow.

Core Mechanism

For a convex cost function $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$, let $\bar{c}$ denote its convex conjugate (Legendre–Fenchel transform):

$$\bar{c}(u) = \sup_x \langle u, x \rangle - c(x).$$

c -Rectified Flow restricts the learned velocity field to the structured form:

$$v_t(x) = \nabla \bar{c}(\nabla f_t(x)),$$

where $f_t : \mathbb{R}^d \rightarrow \mathbb{R}$ is an unconstrained time-dependent scalar potential. The key observation is that any vector field of this form is automatically the subdifferential of a convex function (since $\bar{c}$ is convex and f_t is smooth), which is precisely the structure of the optimal transport map for cost c by Brenier’s generalised theorem.

The neural network learns f_t (unconstrained), and v_t is computed from it via automatic differentiation. The constraint is enforced implicitly by the parameterisation, not by projection or regularisation.

Technical Formulation

Training objective. Given source samples $\{x_0^{(i)}\}_{i=1}^n \sim \pi_0$ and target samples $\{x_1^{(j)}\}_{j=1}^n \sim \pi_1$, paired by a coupling γ_t at interpolation time $t \in [0, 1]$, the c -Rectified

Flow objective is:

$$\mathcal{L}_{c\text{-RF}}(\theta) = \mathbb{E}_{(x_0, x_1) \sim \gamma} \left[\|v_\theta(x_t, t) - \dot{x}_t\|^2 \right], \quad x_t = (1-t)x_0 + tx_1,$$

where $v_\theta = \nabla \bar{c}(\nabla f_\theta)$ is enforced by parameterisation.

Computational guarantee. Let $\mathcal{T}_c(\gamma)$ denote the c -transport cost of coupling γ . Theorem 1 in the paper establishes:

$$\mathcal{T}_c(\gamma^{(k+1)}) \leq \mathcal{T}_c(\gamma^{(k)})$$

for all k , where $\gamma^{(k)}$ is the coupling induced by the k -th iterate of c -rectification. Marginals are preserved exactly at each iteration.

Statistical guarantee. Theorem 2 provides: with n i.i.d. samples from $\pi_0 \times \pi_1$, the estimated c -transport cost satisfies:

$$\mathcal{T}_c(\hat{\gamma}_n) - \mathcal{T}_c(\gamma^*) \leq C_s \cdot n^{-2/(d+4)} \cdot \log n$$

with high probability, where C_s depends on the smoothness class of the optimal transport map and the Lipschitz constant of $\bar{c}$.

Learning or Inference Procedure

Iterative c -rectification:

1. Initialise coupling $\gamma^{(0)} = \pi_0 \otimes \pi_1$ (independent).
2. For $k = 0, 1, 2, \dots$:
 - (a) Pair samples from π_0 and π_1 by simulating the current flow $v^{(k)}$ to obtain coupled pairs $(x_0, x_1^{(k)})$.
 - (b) Train $v^{(k+1)}$ by minimising $\mathcal{L}_{c\text{-RF}}$ on these coupled pairs.

Inference: Simulate the learned ODE $\dot{x}_t = v_\theta^{(K)}(x_t, t)$ from $x_0 \sim \pi_0$ to x_1 using a numerical ODE solver (e.g., Euler with T steps).

What the Guarantee Says

The computational guarantee ensures that repeated c -rectification is a monotone interior algorithm: each iterate reduces the cost without leaving the feasible set of couplings with the correct marginals. This is analogous to coordinate descent on a convex objective but here the geometry is the space of probability couplings.

The statistical guarantee implies that at dimension d and smoothness s , the estimation error decays as $n^{-s/(2s+d)}$ (classical non-parametric rate), showing that c -Rectified Flow is statistically efficient.

However, recovery of the true OT solution requires additional assumptions: connected support of π_0, π_1 and smoothness of the OT map T^* . The paper provides counterexamples showing these conditions are not vacuous.

Experimental Findings

- On synthetic 2-D optimal transport problems with non- L_2 costs (e.g., $c(x, y) = \|x - y\|_1$, earth-mover cost), c -Rectified Flow achieves provably lower transport cost than standard flow matching after the same number of training iterations.
- Monotone cost decrease is confirmed empirically for $K = 1, 2, 3$ iterations of c -rectification across five experimental settings.
- Sample complexity curves match the theoretical $n^{-2/(d+4)}$ rate on synthetic data with known ground-truth OT maps.
- On a molecular conformer generation task with steric clash cost, c -Rectified Flow produces conformers with lower energy than standard flow matching baselines.

Ablations and Interpretation

Effect of $\bar{c}$ parameterisation: Using a learned neural $\bar{c}$ rather than the closed-form convex conjugate produces slightly better results but is harder to optimise and risks losing the theoretical guarantees if the learned function is not truly convex.

Iteration count K : Cost decreases most sharply in the first iteration ($k = 0 \rightarrow 1$) and yields diminishing returns beyond $K = 3$. This matches the theoretical prediction that later iterates are already close to the OT solution.

Disconnect: theory vs. practice: In settings without smooth OT maps (e.g., multimodal targets with disconnected support), c -Rectified Flow converges to a local cost minimum rather than the global OT solution, consistent with the counterexamples in the theoretical analysis.

Reference

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